

Genericity and Modality

Cambridge Handbook of Natural Language Modality

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1 Introduction

What is commonly called “genericity” consists of two distinct classes of phenomena (see Krifka et al. 1995): namely, KIND REFERENCE and GENERIC (CHARACTERIZING) PREDICATION. A sentence with kind reference attributes a property to a kind as a whole, as we see in (1a)/(1b). In contrast, generic sentences, as in (2a)/(2d), generalize over particulars, episodic occurrences and various other states of affairs.

- (1) a. *The potato* / **A potato* is widespread worldwide. KIND REFERENCE
- b. *Potatoes* were introduced into Europe in the 16th century.

- (2) a. *The potato* contains vitamin C. GENERIC PREDICATION
- b. *Birds* have feathers.
- c. *A chocolate bar* melts in the sun.
- d. *Mary* walks to school along the river.

If in a generic sentence, a characterizing property is attributed to a kind, as in (2a)-(2c), this property also holds of its individual specimens, though not necessarily of all of them. For instance, (2b) is true of the kind BIRD by virtue of its subsuming particular members that are individual feathered birds. (2c) is a generic sentence whose subject is an indefinite singular DP; notice that an indefinite singular DP as subject is excluded in kind reference sentences, as in (1a). Example (2d) is

a generic sentence that expresses a characterizing property of an individual. It is true by virtue of particular situations of Mary walking to school, e.g., having observed a number of situations of Mary walking to school, we inductively infer that there is a regularity, which can be captured by (2d). Such generics express habits proper, i.e., regularities of action attributed to ordinary (typically animate) individuals, and are called HABITUALS. They might be best viewed as a subtype of generic predications (e.g., Krifka et al. 1995, Carlson 2011), but some treat them as not prototypical examples of generic sentences, and perhaps even separate from genericity (e.g., Dahl 1995, Nickel 2016).

Modality is an essential property of generic predications, but not of kind reference. All generic predications (including habitual predications) express a regularity, generalization, or pattern that transcends our immediate experiences of the actual world. They do not describe (just) what was or is now, and for which we might have some observable evidence, but also what is possible or likely, what we may predict based on our previous experiences, and possibly also as a matter of a law-like pattern. Their evaluation, therefore, requires access to other possible worlds than the actual one, or any other single possible world.

The most compelling argument for the modal (intensional) nature of generic predication comes from generic sentences that are of a purely hypothetical or dispositional nature, as in (3a) and (3b) below.

- (3) a. The Speaker of the House succeeds the Vice President.
- b. This machine crushes oranges.

Such sentences are true even if they have no verifying instances in the actual world. (3a) is a rule of the United States Constitution that has never been enacted. (3b) can be true even if the machine is still packed in its original box and has not yet been put to use, but was designed for peeling oranges. It can even be true even if the machine only exists as a design.

2 Generic (characterizing) predication

Generic predications are fundamentally opposed to EPISODIC PREDICATIONS. The latter concern contingent, accidental occurrences or states of affairs, involving particular individuals, groups, or situations. They denote singular particular situations, as in (4a)-(4b) or a closed set of particular plural situations, as in (5a)-(5c), which are called ITERATIVE sentences.

- (4) a. Today, we had breakfast already at 7 a.m. SINGULAR SS
b. The delivery man is waiting in front of the house.
- (5) a. We had pizza a few times for dinner last week. ITERATIVE SS
b. During his lifetime Franz met Milena only twice.
c. The red light {repeatedly flashed | flashed on and off} during the fire drill announcement.

The truth value of iterative sentences depends on the occurrence of a closed set of multiple situations, which are not a part of a pattern or regularity. A mere iteration, or plurality of a situation of the same type, is neither sufficient nor necessary for a sentence to be generic.

As many semanticists agree, generic sentences have three hallmark properties (Carlson, 1977): (i) ASPECTUAL STATIVITY, (ii) EXCEPTION-TOLERANCE, and (iii) MODALITY (intensionality). The standard test for STATIVITY is lack of reference to particular situations, evaluable at specific moments of time or locations. For instance, *#Look out of the window, it rains right now* is odd due to the clash between the simple present tense form of the verb *rain*, which is taken to have a default generic interpretation, and the adverbial *right now* referring to a specific time point. In this respect, generic sentences that are based on inherently episodic verbs like *rain* clearly differ from episodic sentences, as in (4a) and (4b). Generic sentences pattern with sentences containing lexically-stative predicates like *(be) tall* or *know French*, which also are not evaluable at specific moments of time and/or locations: e.g., *The runway model was tall and knew French *at 4 p.m. yesterday at the Versace show*. The oddity of such sentences stems from a clash between adverbials referring to specific time points and their main predicates that express properties of individuals that are stable over relatively long intervals, possibly their whole lifetime. To be evaluated with respect to a single temporal or spatial location is systematically uninformative and hence pragmatically odd.

The second property of generic sentences, EXCEPTION TOLERANCE, is one of their most intriguing features. While there are generics permitting no exceptions whatsoever under any conditions, e.g., universal laws like *The Earth revolves around the Sun* or definitional statements based on essential properties like *A madrigal is polyphonic*, many generic sentences allow for exceptions, while still remaining true. There is an extensive body of studies on the exception tolerance of generics in semantics and also in related fields of cognitive science. The extant proposals include analyses in terms of normality, typicality, stereotype, default logic, or some probabilistic idea (for overviews see Krifka et al. 1995, Pelletier and Asher 1997, Nickel 2017). Different analyses tend to focus on some facet(s) of exception-tolerance, and we now have valuable insights into different types and

numbers of tolerable exceptions, and also counterexamples that invalidate generic sentences. However, there is, as yet, no comprehensive account of this key property of generic sentences, starting with the fundamental observations that different types of generic sentences sanction different types of exceptions and a different number of exceptions. Some generics may be true based on majority satisfaction, e.g., a majority of members of the kind satisfying the requisite characterizing property:

(6) Ravens are black. / Dogs bark.

However, there are generics that are judged false or unacceptable despite majority satisfaction:

- (7) a. Supreme Court judges have a prime Social Security Number.
- b. People are over three years old.
- c. Prime numbers are odd.
- d. Crocodiles die before they attain an age of two weeks.

For example, (7a) is strange, or even false, as a generic sentence, even if all judges happen to have an odd SSN, because there is no plausible causal factor or rule that would back up the property of having an odd SSN as a characteristic property of Supreme Court judges across different possible worlds (see e.g., Nickel, 2017).

Neither is majority satisfaction necessary for the truth of all exception-tolerating generics. What is treated as a characterizing property of a class need not be prevalent among its members. It is sufficient that it is a particularly salient, subjectively impressive, memorable, or striking feature of some of its members that distinguishes it from related kinds. For instance, (8a) holds in spite of the fact that only adult male lions have a mane, which is a minority among lions. Moreover, some generics are true by virtue of a negligible minority exhibiting the requisite characterizing property. (8b) is true even if only about 1% of mosquitoes carry the West Nile virus.

- (8) a. Lions have a mane, though cubs and female lions do not.
- b. Mosquitoes carry the West Nile virus, though 99% do not.

This has raised the interest of cognitive scientists in such sentences (e.g., Leslie 2007a, 2008, see also further below). However, it is not entirely clear whether and how a variety of subjective, psychological factors can be accommodated within a truth-conditional account of generic sentences (see Nickel 2017, Ralston 2024) and what kind of modality they involve.

Third, generic sentences have an essentially MODAL (intensional) component (as originally proposed by Lawler 1972, 1973, Dahl 1975) (see 1.1 Conceptual foundations, 1.2 From modal logic to modal semantics). Their truth conditions involve modal quantification over other possible worlds than just the actual world, namely those worlds where these patterns, regularities, and law-like rules also hold. This quintessentially modal nature of generic sentences makes their meaning irreducible to some notion of quantity, plurality of individuals and/or cases, or to some quantity-based or statistical criterion.

In support of this point, we observe that a closed set of individuals and/or cases that hold at a given world/time is neither sufficient nor necessary for the truth of generic sentences. It is not necessary, as generics with no verifying instances in the actual world can still be true (see (3a)/(3b)). It is not sufficient either, even if the relevant closed set of particulars or samples that form the base of the generalization at a given world/time is very large. Suppose there is a very large ceramic urn containing a million marbles. You take out a marble a number of times, and each time the marble you take is black. You might infer that all, most or many of the marbles in this urn are black. And yet, it would be odd to describe what you concluded with *Marbles in this urn are black*, which contains a bare plural subject, and is naturally understood as a generic statement. This is because your observations and conclusions concern the specific, delimited set of marbles in this particular urn. A better description of your observations would be by means of an episodic sentence with the definite plural subject, e.g., *The marbles in this urn are black*.

For generic sentences to be true and appropriate, it is not sufficient to express contingent facts about a closed set of individuals or situations at a given actual world and time; rather, they express generalizations that extend to other, hypothetical worlds and times. Consequently, they are modal statements (see 2.5. The diversity condition across modal flavors). Particular samples, instances, states of affairs that we may observe in the actual world matter for the evaluation of a generic sentence insofar as they count as evidence for the underlying rules or causes. As many agree, the notion of causation, a modal notion, is relevant for motivating the truth conditions of generic sentences, and generic sentences support counterfactual inferences (Lewis, 1973).

The modality of generic sentences separates them from sentences with *extensional* quantifiers like *all*, *every*, *most*, *many*, *some*, and also from distributive predications about sum individuals. Consider the following contrast (adapted from Pelletier and Asher 1997, p. 1132-3):

- (9) Suppose a world where the remaining pandas happen to have only three legs.
 - a. All pandas have three legs.

- b. Pandas have three legs.
- c. Pandas have four legs.

Explicitly quantified sentences like (9a) may have either an extensional or a generic (intensional) interpretation. Sentences with bare plural subjects like (9b) and (9c) are naturally understood as generic. On a purely extensional level, (9a) asserts that all pandas have three legs, which happens to be true in the world of evaluation. But it does not express a generic statement, because it makes a purely quantitative claim about a closed set of (existing) pandas in the given domain of quantification.

Interpreted generally without reference to a given set of pandas that happen to have three legs, the quantified sentence (9a) is false, just as (9b). In contrast, (9c) is true, even if in our particular world of evaluation, no pandas have four legs. This is because for the truth value judgments of generic sentences, the actual extension of their subject argument at a particular possible world does not matter, or it may matter only insofar as it provides evidence for some underlying factors or causes. On their generic interpretation, (9a)-(9c) make claims about an open set of (realistically) possible pandas, also in other (possibly all) accessible possible worlds. These claims are true or false based on the PANDA-kind specific biological laws, which function as causal factors determining the development and properties of pandas. This is reflected in our explanatory reasoning, as in *This animal has four legs, because it is a panda.*

In sum, the property of exception-tolerance and the quintessentially modal nature of generic sentences make implausible any reductionist analysis that relies on some quantity-based or statistical criterion, a single quantificational operator or an expression of quantity. Moreover, these two properties, individually and jointly, also indicate that while some notion of quantity, plurality, statistical support, probability, or proportion may be relevant for the truth-conditions of many generic sentences, they are neither necessary (see (3a)/(3b)) nor sufficient (see (5c)/(5b), (7)) for their truth. The source of these properties lies in the law-like modalities of generic sentences to which we turn next.

3 Varieties of generic predications

3.1 Three kinds of generics

The modal nature of generic sentences was emphasized by Dahl (1975), who argued that they express law-like (or NOMIC statements. Dahl distinguished between “DESCRIPTIVE nomic statements”, which express laws (e.g., physical, biological) about what holds in the physically possible worlds, and “NORMATIVE nomic statements”, which express social (moral) norms, customs, or regulations that hold

in the morally perfect worlds. Carlson (1995) outlines two main perspectives on generic statements. One is characterized in terms of an “INDUCTIVE model”, on which they express inductive generalizations that are based on (un)observed instances from which we infer general rules, such as *Pigeons sit on statues*. The other is a “RULES-AND-REGULATIONS model” for which rule-governed activities such as game rules like *Bishops move diagonally* provide the best examples. Carlson argues that these models fit different types of generic sentences.

Carlson’s distinction is broader than Dahl’s distinction. Carlson also allows for mixed cases, namely generics that combine the features of the inductive and rules-and-regulations model, which results in three broad classes of generics. First, some generics express purely descriptive generalizations that require some verifying instances in the actual world to be true, and hence for them only the inductive model is appropriate. The second broad class comprises generics that are true by virtue of rules, norms, and social agreements of various types that we may know directly without empirical observations of the physical world. The third class of generics is based on observed instances but results in rules as well; they correspond to Dahl’s descriptive nomic statements. Let us now address each class in turn.

3.2 Generalizations based on induction

Generic sentences that express inductive generalizations can be judged true merely by virtue of some sufficiently large set of observed instances and occurrences of the actual world, provided that we detect an underlying pattern that we assume will apply to additional instances in the future (see 5.1. Modality and tense, 5.3. Modality and the future), or in worlds that are similar to ours. According to Cohen (2012), who focuses on inductive generalizations, generic sentences have a “uniform” ordering source: they presuppose that the past, from which the observations are drawn, is similar to the future. In this respect, generic sentences make claims that transcend our immediate experiences of the actual world. However, they often do not depend on some easily identifiable causal relation or explanation, laws or norms, and so are not amenable to causal or normative reasoning. The following examples illustrate this:

- (10) a. Pigeons sit on statues.
 # This animal sits on statues, *because* it is a pigeon. (cause)
 # If this animal is a pigeon, it *should* sit on statues. (norm)
- b. Barns are red. (A statement about New England).
 # This building is red, *because* it is a barn. (cause)
 # If this building is a barn, it *should* be red. (norm)

While the truth of descriptive generics that best fit the inductive model depends on some sufficiently large sample of relevant instances, a mere notion of quantity, frequency, or probability that can be fixed once and for all is not enough to ground their truth. Their truth also depends on other factors that influence why and how we infer generalizations from observations of a few samples. What matters is how the relevant samples that underlie the generalization come to our attention, rather than some mere statistical correlation (as we have already seen in (7a)/(7d)), frequency, probability, or proportion. It also matters whether the characterizing property has a utilitarian purpose and facilitates the affordances of members of a kind. Consider the following contrast:

- (11) a. #Cars have black tires. Carlson 1977, 2009
 b. Cars have tires.
 c. Cars have air-filled tires.

Even if most car tires are black, (11a) is odd, because the blackness of tires is irrelevant to what is necessary to their being cars, to their proper functioning as cars. In contrast, having tires, and for the tires being air-filled, is necessary for cars to function properly, which motivates the felicity of (11b) and (11c). Put in the terms of Prasada and Dillingham (2006), blackness as a property of tires is a *t*-property (“statistical” property), while air-filled as a property of tires, most people might judge as a *k*-property (“principled” property).

For a generalization to be true (or false) it matters how exactly exceptions to it are judged as failing to satisfy its characterizing property, as the contrast between (12a) and (12b) shows:

- (12) a. Barns are red. TRUE
 b. Books are paperback. FALSE

(12a) is true by virtue of the majority of barns in New England being painted red, and it is not falsified by exceptional barns that are painted a different color. In contrast, (12b) is false, even if the majority of printed books are paperbacks, and it is falsified by the existence of hardbacks. We are likely to judge a generic sentence to be true, according to Leslie (2008), if exceptions to it constitute “negative counterinstances”. Barns that are not painted red constitute negative counterinstances to red barns, because they simply lack the positive characterizing property without providing any alternative positive property that would be equally (subjectively) salient or striking as being painted red. In contrast, in (12b), hardbacks are viewed as “positive counterinstances” to paperbacks, because they have an alternative positive property that cannot be ignored. We can even see paperbacks and hardbacks as two subkinds of (physical) books.

Inductive generalizations are inherent to sentences like (13), which are called “habitual”, in the strict sense of the word “habit” (regularity of action attributed to an ordinary individual, typically animate, and human).

(13) John drinks beer.

As observed by Lawler (1972, 1973), such sentences can be understood in two ways, which he termed “universal” and “existential”. To avoid the connotation of the first term as lacking exceptions, we will refer to these readings as “strong” and “weak”. On its strong reading, (13) means that John is a beer drinker, or that he drinks beer with some regularity. Evidence for this would be situations of John’s drinking beer, allowing for exceptional cases in which John could drink beer, but in fact does not. On its weak reading, (13) means that John is not averse to drinking beer, and may drink beer in an appropriate situation in which beer is offered. Such judgments can be based on evidence that John has drunk beer only once in the past, and on the lack of evidence that he would refuse to drink beer in the future. Notably, (13) has a modal component as part of its meaning in both cases, and the distinction between the strong reading and the weak reading is reminiscent of the distinction between strong modals like *must* and also to weak modals like *might*, which have close affinities to ability modality (see 2.3. Circumstantial and ability modals).

Habitual sentences like (13), and generic sentences that are headed by generic predicates derived from inherently episodic predicates, are grounded in situations that can be described by these episodic predicates. In English, the simple (non-progressive) present tense form has a predominantly habitual/generic reading, whereas the progressive form predominantly denotes a stage of a larger eventuality ongoing at a reference time (e.g., *John is drinking beer*). In contrast to generic predicates that are derived from episodic predicates, lexically stative predicates like *know French* denote characterizing properties that are grounded in specific actions (e.g., speaking and writing French, or reacting appropriately when listening to French). But they have no episodic counterparts that could be used to describe such particular actions, counting as evidence for their truth. (See Chierchia 1995 for such “inherent generics”, a notion that is sometimes extended to all stative predicates.)

3.3 Generalizations based on rules and regulations

Some of the most compelling examples of generic statements are those that are not derived inductively from observations but express rules. For them, only the rules-and-regulations model of genericity (Carlson 1995) is appropriate. They create new forms of reality or make possible new forms of behavior that cannot exist without

the rules. In this way, they contribute to the structuring of the social reality, and we may learn them directly, independently of any empirical observations of the physical, actual world. Good examples are constitutive rules (in the sense of Searle 1995), such as rules of chess:

(14) In chess, bishops move diagonally.

There are also constitutive rules that define purely hypothetical actions for which there need not be any verifying instance in the actual world, such as (3a) or (15).

(15) Sam handles mail from Antarctica.

Other generics that express constitutive rules are those that mainly concern conditions on (the appropriate use of) language. They serve to introduce a new term or restrict the interpretation of an existing term, as in the following examples:

- (16) a. A heptagon has seven sides.
b. An even number is the sum of two prime or two even numbers.

Analytic truths like (16a) are true by virtue of the meaning of the words they contain. The truth of a theorem or conjecture, as in (16b), is or is suspected to be a logical consequence of the axioms, which are definitions, and previously proved theorems, and hence such sentences are analytic statements as well. Lawler (1973) and Burton-Roberts (1977) observe that they often refer to the class of entities by singular indefinites, even though bare plurals are possible as well, as in *Heptagons have seven sides*. In contrast, singular indefinite subjects in generic sentences that are based on induction and lack a discernible cause or norm are odd, e.g. *#A barn is red*. In general, singular indefinite generics suggest that the predication expresses an essential property (Lawler, 1973; Cohen, 2001), involves causation (Carlson, 2009), or concerns a property that entities have “in virtue of” belonging to a given class (Greenberg, 2003). It has also been observed that they have a moral or prescriptive quality (Corblin, 1987; Mari, 2008; Cohen, 2022).

The modality of the above generics is akin to that of deontic modality, given that they express rules or norms about what ought to be the case, e.g., in compliance with the U.S. Constitution, some job description, or how the terms *natural number*, *heptagon* or *madrigal* ought to be understood and used. Deontic modality is relevant for another type of generics for which only the rules-and-regulations model of genericity is plausible, namely regulative rules (in the sense of Searle 1995). They regulate behavior that exists independently of the rules or norms they specify. Paradigm examples are rules of the road or normative rules of etiquette:

- (17) a. In Great Britain, one drives on the left side of the road.
 b. A gentleman opens the door for a lady.

Having a deontic character, they can be rephrased as imperatives (e.g., *Drive on the right!*). As with constitutive rules, if regulative rules have exceptions, they must be explicitly stated. Regulative rules may also intersect with normative rules, as illustrated by (17b). Normative rules concern standards of behavior or beliefs about what is viewed as ideals to be aspired to, desirable, or enforced by social pressure within a social group, or society at large. The word *gentleman* in (17b) adds to the regulative rule of etiquette the association with an ideal to strive for by virtue of being a positive evaluative term.

Another type of generics that can be subsumed under the rules-and-regulations model are those that express conventions, customs, or stereotypes such as (18).

- (18) A boy doesn't cry.

This is a normative statement that can be expressed by a deontic modal, as in *A boy must not cry*. However, such sentences also express an exclusion criterion from a positively charged concept: If you cry, you are not a *real* boy.

Generic sentences that best fit the rules-and-regulations model of genericity provide compelling evidence that a quantity- or statistically-based account for all generic sentences is implausible. What counts as evidence for their truth does not depend on particular occurrences in the actual world, but rather on whether the criteria imposed by the rules are met in some possible (ideal) world, and possibly in all such worlds.

3.4 Generalizations based on a hybrid model of genericity

There are generic sentences that combine elements from both the inductive model and the rules-and-regulations model of genericity. They express generalizations we infer from empirical observations of the real world and lead to the recognition of underlying laws, causal factors, or guiding principles. These laws might hold without exception, like the universal laws of physics, or might allow for exceptions, such as laws of biology, psychology, geography, and economics. What we can observe or experience in the real world, the quantitative or statistical evidence we may extract from it, is only relevant for its truth insofar as it is evidence for such underlying laws. Consequently, their truth conditions and modality are more complex than those of descriptive generics that only rely on empirical observations from which we infer an underlying pattern, as well as truth conditions of generics that impose rules, regulations, or norms on the world. Generic sentences of this type can be grounded in universal laws of nature, as in (19a) and (19b) below,

and as such are necessarily true in all possible worlds for which the laws of nature hold.

- (19) a. Oil floats on water.
b. If you drop an apple, it falls to the ground.

Having observed oil floating on water and apples falling from trees to the ground on a sufficient number of occasions, and no counterexamples to these observations, we may infer (19a) and (19b), respectively. We may assume that the events we observe follow natural laws, which fall under the rules-and-regulations model of generics.

We find this combination of the induction model and the rules-and-regulations model also with generic sentences that involve definitions. Krifka (2012) illustrates this with the following example:

- (20) A donkey has 62 chromosomes.

On the one hand, it is an empirical observation that donkeys have 62 chromosomes (compared with horses, which have 64, and mules, which have 63). Hence, the truth of (20) is based on induction. On the other hand, the number of chromosomes has come to be one of the defining properties of a biological kind. Hence, (20) has the status of a DEFINITIONAL generic, and fits the rules-and-regulations model. Following an argument in Kripke (1980), sentences like (20) are analytical statements (based on rules and regulations) that are at the same time a posteriori (based on observation). Notice, also, that such sentences can be naturally expressed with indefinite generic subjects, which could be seen as a reflection of their analytical nature (see Burton-Roberts 1977).

4 Modal approaches to genericity

4.1 The modal nature of the generic quantifier

Dominant approaches to genericity and habituality that use some concept of modality can be traced to the 1970s. For example, Lawler (1973) observed that generics can be expressed with modals like *must* and *can*, and Dahl (1975) assumed a null generic operator GEN that he described as quantifying over a specific set of possible worlds, e.g., the worlds where the law expressed by the generic holds without exception, just as modal operators do. Modality is generally divided into two large categories, epistemic modality (see 2.1. Epistemic modality) and root or “circumstantial” modality (Kratzer, 2012) (see 2.3. Circumstantial and ability

modals). It is evident that the modality of genericity is of the second type. It does not qualify the epistemic status of a proposition, i.e., whether it is considered certain, probable, or possible, but rather, it is part of the truth conditions of the proposition itself. Circumstantial modality, in turn, has been divided into priority modals like deontic, bouletic, and teleological modals, on the one hand, and dynamic modals, on the other hand, which comprise ability modals and intrinsic modals (see Portner 2018, also 1.1. Conceptual foundations).

However, the modality of generic predications cannot be reduced to the modal notions that have been independently proposed for other linguistic phenomena than generic predications. While the truth conditions of generic predications interact with or might be akin to deontic, epistemic, and bouletic modality (see 1.1. Conceptual foundations, 2.1. Epistemic modality), for instance, they cannot be reduced to such modalities, although they can be in the scope of deontic, epistemic, and bouletic modals. This is illustrated by the following examples:

- (21) a. Boas presumably lay eggs, as most snakes do.
 b. I wish you would keep your room tidy.

Neither can the modal import of generics be reduced to or viewed as akin to the modality (intensionality) independently proposed for the analysis of tense, imperfective aspect or progressive aspect as one of its special cases. Other known modal notions like essence, which are discussed in the epistemology of modality, are also inadequate.

We may also mention that there are problems with modeling generic predications as modals, understood as quantification over possible worlds. Krifka (1995) observes that the sentence *A turtle is long-lived* cannot be expressed as: ‘In all worlds in which the biological predispositions of turtles are realized, if x is a turtle, x is long-lived’. This would result in highly non-normal worlds full of turtles, as they all reach maturity. One possibility is to allow generic modals to quantify over situations, that is, limited worlds. For example, when considering individual turtles x and the local situations in which they exist, we conclude that in ideal situations in which their genetic traits are expressed, they are long-lived.

Modal operators come in strong and weak varieties. For example, *must* as a deontic modal expresses an obligation, and *can* a permission. We have seen in (13) that the null generic operator GEN allows for a strong and a weak reading as well, without any overt marking of either. In place of the null generic operator GEN we can have explicit ADVERBS OF QUANTIFICATION like *always*, *usually*, *generally*, *typically*, *often*, *sometimes*, *rarely* and *never* which correspond to modal quantifiers of different strength, like *necessarily*, *certainly*, *probably* and *possibly* (see e.g., Farkas and Sugioka 1983, among others). It should be stressed that sentences with adverbs of quantification can have either an episodic interpretation

(see (22a)) or a generic one (see (22b)). Adverbs of quantification do not ‘mark’ or enforce a generic interpretation of sentences (see e.g., Pelletier and Asher 1997).

- (22) a. Last week, the children in this school were often/seldom silent and shy.
b. Children in this school are often/seldom silent and shy.

4.2 The marking of generic sentences

While modality is typically marked in some way, for instance, by modal verbs, adverbs, particles, or markers of grammatical mood, generic predications often are formally unmarked. In fact, they often use the least marked form of the tense and aspect system in a given language (Dahl, 1995), such as the simple present in English, as in *Dogs bark* (on its generic reading). There is a widespread, long-standing, but not uncontroversial, belief that natural languages have no overt, grammatical markers of genericity (see e.g., Leslie 2008, Leslie and Gelmann 2012, Johnston and Leslie 2012, Rhodes et al. 2018, Cohen 2022, Schiller 2023, i.a.). Hence, at first sight genericity appears to be endowed with modality which is quite unlike other types of modality.

However, in a number of languages, generic predications can be encoded by dedicated morphological markers or syntactic constructions, often called “habitual” (see e.g., Dahl 1985, 1995), which enforce only readings having to do with regularities, habits or rules of various types, and exclude any episodic readings. Some examples are the English *used to* construction (which only occurs in the past, perhaps because the simple present tense in English has a predominantly generic/habitual meaning), the German *pfl egt zu* ‘to tend to’ construction, which can be used in all the tenses, presumably because German present tense verbs can be used in either generic or episodic sentences, the Hebrew generic marker *Pron* (Greenberg, 1998), the *be* construction in African American English (Green, 2000) and the Hebrew construction employing the root *hyy* ‘be’ in the past tense combined with the main verb in its participial form (Boneh and Doron, 2010, 2013), the Swahili prefix *hu-* (Dahl, 1985; Krifka et al., 1995), the generic/habitual morpheme *-va-* in West Slavic languages Czech, Slovak and Polish (Filip, 1993, 1994; Filip and Carlson, 1997; Filip, 2026), and the Tlingit “habitual” morphemes (Cable, 2022).

What these overt markers have in common is that they are not obligatory for the expression of habitual or generic statements, and require that there be verifying instances in the actual world for the expressed regularity. The latter property is known (perhaps somewhat misleadingly) as the ‘actualization entailment’. To illustrate this property, consider the following Czech example:

- (23) *Scenario*: We have just bought a new machine for crushing oranges, but we have not yet used it, and it is still packed in its original box.

Tento stroj drtí pomeranče. (TRUE) / # drtívá pomeranče. (FALSE)
 this machine crush.3SG.PRS oranges / # crush.GEN.3SG.PRS oranges
 (Intended:) ‘This machine crushes oranges.’

In the given scenario, only the verb that is unmarked for genericity, a simple (primary) imperfective verb, is acceptable, and (23) is true by virtue of the machine’s design, giving rise to the relevant counterfactual inferences. The verb with the morpheme *-va-*, which is not to be confused with the imperfectivizing suffix (see (Filip, 1993, 1994; Filip and Carlson, 1997; Filip, 2026)), is unacceptable in the given scenario, because it presupposes that situations of crushing of oranges actually occurred. At the same time, its truth also depends on the machine’s design, and sanctions the appropriate counterfactual inferences that are the hallmark property of generic statements. This is also one of the reasons why the morpheme *-va-* is treated as a generic marker *sui generis*, and so is glossed as GEN (see Filip and Carlson 1997 and Filip (2026), *pace* Dahl 1995, i.a.).

Most importantly, the behavior of the Czech generic and habitual marker *-va-* illustrates the following general point (see Boneh and Doron 2010; Filip 2026): namely, the actualization inference that is tied to overt so-called “habitual” markers does not mean that sentences in which they occur lack a modal component, and neither does it follow that they, therefore, are merely markers that function as extensional quantifiers over situations, which do not qualify as markers of genericity, but rather are merely tense and/or aspect “habitual” markers (*pace* Dahl 1985, 1995; Cable 2022, i.a.). That the actualization inference is not necessarily correlated with lack of modality should not be surprising, given that there are modal operators that have the actualization requirement (Bhatt 1999, Hacquard 2006).

Apart from modality (intensionality), there are two other properties that the Czech marker *-va-* shares with the null generic operator GEN. First, *-va-* introduces a quantifier which just like GEN quantifies over classes of individuals only, i.e., does not quantify over episodic situations (*pace* Dahl 1995):

- (24) Vrány bývají černé.
 ravens be.GEN.3PL.PRS black
 ‘Normally/Generally ravens are black.’

Second, while *-va-* is often used to express inductive generalizations, its use is *not* restricted to generalizations that only fit the inductive model (see Section 3.2, and Filip 2026). It may also be used in generics best fitting the rules-and-regulations model.

There are also two other key properties that separate the Czech marker *-va-* from GEN: namely, *-va*-marked generics require that there be instances in the actual world that serve as evidence for the expressed generalization, just as so-called “habitual” markers in other languages do, and also (at least possible) exceptions or counterexamples to the expressed generalization. A possible motivation for the co-occurrence of these two requirements is that the reasoning with exceptions involved in *-va*-marked sentences is based on the speaker’s more or less direct experience of the facts, particulars in the actual world that provide evidence for their truth (or falsity). Put differently, the actualization inference of the Czech *-va-* is a precondition for the requisite reasoning with exceptions. It is an open empirical question whether “habitual” markers and generic markers in other languages that introduce the actualization inference also require that there be (possible) exceptions or counterexamples to the generalization expressed by sentences in which they occur.

A final point to be made here concerns the strong/weak readings of habituais and generics. They are discussed above in connection with (13), which is an example of a habitual in the strict sense, and inductive in nature in both readings, as it is fundamentally based only on observations. In Czech, a comparable type of predication can be expressed by means of a *-va*-marked verb and the corresponding unmarked verb:

- (25) a. Jágř hraje^{IPFV} hokej.
 Jágř play.3SG.PRS hockey
 ‘Jágř plays hockey.’
 b. Jágř hrává hokej.
 Jágř play.GEN.3SG.PRS hockey
 ‘Jágř plays hockey / tends to play hockey.’ I.e., as a rule, often, seldom.

(25a) with the verb that is unmarked for genericity, which is imperfective (and also formally unmarked for imperfectivity), allows for either reading, in parallel to (13). On its weak reading, (25a) can be satisfied by just one verifying episode, meaning that Jágř has an ability or is not averse to play hockey. On its strong reading, (25a) is satisfied by a sufficiently large iteration of situations of hockey playing, but not by a single situation, and it can also be used to imply that Jágř is a hockey player by profession. The generic morpheme *-va-* restricts the interpretation of (25b) to only a strong reading, but unlike (25a), (25b) would not be used to convey that Jágř is a hockey player by profession (Filip, 2026).

4.3 Genericity analyzed with a bipartite modal operator

One feature of generic statements that sets them apart from simple modal statements is that they are formed by a BIPARTITE OPERATOR. Heim (1982) observed that generic statements are similar to conditional clauses. One of her examples is *A cat that has been exposed to 2,4-D goes blind*, which she interprets similarly to *If a cat has been exposed to 2,4-D, it goes blind*. Carlson (1989) also argues for a bipartite operator, as he observes two readings for sentences like (26), which can be analyzed as indicated.

(26) Typhoons arise in the Pacific.

- a. ‘If typhoons arise, it is in the Pacific.’
 $\text{GEN}(x: \text{ a typhoon; } \exists s[s: \text{ a situation in the Pacific, } y \text{ arises in } s])$
- b. ‘In the Pacific, there arise typhoons.’
 $\text{GEN}(s: \text{ a situation in the Pacific; } \exists x[x: \text{ a typhoon, } x \text{ arises in } s])$

For the logic of the null generic operator GEN, this means that we have to assume a structure with a restrictor and a matrix (see also Schubert and Pelletier 1989; Krifka et al. 1995). As a modal operator, GEN in (26a) quantifies over typhoons, its variable x introduced in the restrictor. Assuming world and time indices i that behave according to the natural laws, the formula states that if x is a typhoon at i , then x arises in a situation s in the Pacific. Note that there can be more than one variable bound by GEN in the restrictor, as in *Lizards and snakes attack each other*, which can be analyzed as $\text{GEN}[x \text{ is a lizard, } y \text{ is a snake, } s \text{ is a situation that contains } x \text{ and } y; x \text{ and } y \text{ attack each other in } s]$.

The bipartite structure of the generic predication is often not visible in the overt syntactic structure, as the restrictor can be specified implicitly, and is dependent on the information structural properties indicated by prosody (see e.g., Sgall et al. 1986, Partee 1991, von Stechow 1994, Krifka et al. 1995). For (26), the reading corresponding to (26a) is facilitated with focus on *in the Pacific*, while (26b) assumes focus on *typhoons arise*. There are various proposals how information structure can result in different mappings of semantic material to the restrictor or the nuclear scope of the generic quantifier. One prominent approach is alternative semantics, which assumes an existential closure over the focus alternatives that defines the restrictor (Rooth 1985). For example, in *OFFICERS escort ballerinas*, focus on *officers* leads to the interpretation $\text{GEN}(x, s: \exists y[x: \text{ alternative to officers, } x \text{ escort } y \text{ in } s, y: \text{ ballerinas}]; x, s: \exists y[x: \text{ officer, } x \text{ escort } y \text{ in } s, y: \text{ ballerinas}])$, ‘whenever persons escort ballerinas, officers escort ballerinas’. Alternatively, it has been proposed that the restrictor is defined by given material (Krifka, 2001).

Languages differ in their morphosyntactic requirements to express this bipartite structure (see 1.5. Empirical foundations of modality: expressing modality

across languages). For example, imperfectives in Italian and French are taken to be odd out-of-the-blue, and require either an adverb of quantification, a *when*-clause, or a conversational background which provides some salient restrictor to facilitate generic or habitual reading (Bertinetto, 1994; Bonomi, 1997; Bertinetto and Delfitto, 1996). To take an example from French, generics with singular indefinite subjects are odd, unless some overt restrictor is provided:

- (27) a. # Un chien mange des croquettes. Mari 2008
 ‘A dog eats croquettes.’
 b. En Autriche, un chien mange des croquettes.
 ‘In Austria, a dog eats croquettes’

In Czech, generics that are marked by *-va-*, in contrast to generics without it, are odd in out-of-the-blue contexts, as well, but adding an appropriate restrictor renders its use felicitous.

- (28) a. Psi štěkají^{IPFV}. Filip 2026
 dogs bark.3PL.PRS
 ‘Dogs bark.’
 b. # Psi štěká**va**jí.
 dogs bark.**GEN**.3PL.PRS
 ‘Dogs (tend to) bark as a rule.’ / ‘Normally, dogs bark.’
 c. Psi štěká**va**jí, s výjimkou některých druhů psů jako chrti.
 dogs bark.**GEN**.3PL.PRS with exception some breeds dogs.PL.GEN like greyhounds
 ‘Dogs tend to bark, except for some dog breeds like greyhounds.’

The presence of a bipartite structure may be a reason why the null generic operator GEN itself is often not indicated overtly. If the bipartition is marked by information structure, then this can provide sufficient clues for the presence of an operator that needs this structure for interpretation.

5 Other analyses of genericity

In this final section, we will discuss approaches to genericity, including habituality, that either do not propose a modal analysis or reduce its modality to modal notions that have been independently proposed for other linguistic phenomena (e.g., progressive aspect, modal verbs), and argue that they are problematic.

5.1 Genericity as default reasoning

One prominent approach to generic sentences analyzes them as inferences that allow for exceptions. To use a commonly cited example, *Birds fly* can be understood as follows: ‘Whenever there is a bird x , we can infer that x flies (i.e. that x has the ability of flying).’ There might be exceptions to this rule, e.g. if it happens that x is a penguin; hence, the inference is typically a default rule that holds for “normal” cases. Several formal approaches have been developed, such as DEFAULT LOGIC (Reiter, 1980) and circumscription (McCarthy, 1980). Such logics allow that the inference from *Tweety is a bird* to *Tweety flies* can be drawn if it is not known that Tweety is a penguin, or another non-flying bird; in case this becomes established later, the conclusion can be withdrawn.

As Krifka et al. (1995) observe, such default inferences typically are not expressions “in” a language, but “about” a language. Hence, they are typically not understood as having a truth value, or as being propositions (understood as functions from world-time indices to truth values). This is problematic, as generic clauses can be embedded in larger clauses, as in *Animals that fly have wings*. According to standard theories of conditionals, the relative clause (*animals*) *that fly*, itself a generic statement, should contribute a proposition as a restrictor for the head noun, which then is part of a generic predication, *have wings*. Within the model of truth-conditional semantics, generics should clearly have truth values.

In the psychological literature, it was also proposed that generic sentences represent a default mode of generalizing, different from the quantificational mode, as in (Leslie, 2007b). Against this view, it has been argued that quantificational approaches can handle exception tolerance by restricting the domain of quantification (Lazaridou-Chatzigoga et al., 2023). It is remarkable that, even though it is disputed whether all languages have numerals and nominal or adverbial quantifiers, that all languages have some means of expressing generic (and habitual) statements is not disputed.

5.2 Genericity as a tense or aspect

Generic and habitual interpretations of sentences are indirectly reflected in the choices of grammatical markers of tense, aspect and mood (TMA), given that they have semantic affinities to different categories of the TMA system (see 4.2. Mood and Modality, 5.1. Modality and tense, 5.2. Modality and aspect, 5.3. Modality and the future). This motivates the often held assumption that genericity and habituality are to be subsumed under one of the TMA categories, most commonly IMPERFECTIVE ASPECT (“habitual aspect”) or tense (“generic tense” or “(g)nomtic tense”), rather than being regarded as a grammatical category in its own right.

The most widespread claim is that genericity and habituality fall under im-

perfective aspect, which was popularized by Comrie (1976, p.25ff.). Comrie splits imperfectivity in natural languages into a ‘habitual’ and ‘continuous’ subdomain, where habituality covers all the uses of imperfective forms that are not episodic. Subsequently, habituality and genericity are often grouped together as one class, and directly opposed to progressivity under the umbrella of imperfectivity, even if imperfective forms have additional interpretations such as counterfactuality (Hacquard, 2006). On this view, the main goal is to identify the common semantic core of genericity/habituality and progressivity, from which additional constraints can systematically derive their differences.

One common strategy is to assume that genericity/habituality and progressivity share or intersect in their modal component. Progressives have been analyzed as modals, due to the so-called imperfective paradox (Dowty, 1977; Landman, 1992; Portner, 1998), according to which a sentence like *John is crossing the street* can be true even if John never manages to complete the crossing. Proponents of a common modal analysis of generics/habituals and progressives include (Bonomi, 1997; Cipria and Roberts, 2000; Lenci and Bertinetto, 2000; Menéndez-Benito, 2002; Ferreira, 2005a, 2016; Deo, 2009; Hacquard, 2009; Anand and Hacquard, 2009). For instance, Ferreira proposes that the imperfective operator has a singular and a plural variant, and both share the same modal quantification over possible worlds, and the same temporal component, represented by means of the existential quantification over eventualities, namely, singular eventualities in the case of the progressive reading and plural eventualities in the case of generic/habitual readings. In a progressive sentence like *John is playing soccer*, the imperfective form refers to a stage (in the sense of Landman 1992) of a single larger eventuality e in the actual world w^* (temporal component), which will continue in its future continuation branches and culminate in case all potential external obstacles are removed (modal component). Similarly, a habitual sentence like *John plays soccer* is true in all the worlds in which there were multiple eventualities of John playing soccer before the utterance time, and there are ongoing multiple eventualities of John playing soccer (temporal component), whereby all potential external obstacles to its continuation (e.g., by breaking one’s leg) are removed (modal component). Consequently, its truth requires that a future multitude of soccer-playing eventualities be guaranteed.

At first blush, the claim that the modality of genericity/habituality can be reduced to the modality of progressivity seems plausible, especially as progressive forms do not exclude habitual readings (see e.g., *Gabriele is drawing portraits (these days)*, *Jim is always losing his glasses*). However, this way of thinking about the modality of genericity/habituality cannot (adequately) accommodate even some of the core data. First, the assumption that “habituals are about plural events” (Ferreira, 2016, p.353) does not cover generics like *This machine crushes*

oranges on their purely intensional reading. Acknowledging such data, Ferreira (2005b, p.120) suggests that they have a covert modal similar to *can* in their logical structure. In fact, his analysis only covers some generics that fit the inductive model (Section 3.2), but not generics that presupposes some underlying rule, principle or cause, which fall under the rules-and-regulations model. Second, even paradigm examples of habituais like *John plays soccer*, which are paradigm examples of generics that fit the inductive model, and on their weak reading need not be satisfied by multiple events (see also above the example (13)). For *John plays soccer* to be true it is sufficient that John has played soccer only once, and it may not even be guaranteed that he will play soccer in the future (see e.g., Hacquard 2006, p.81).

Instead of trying to identify a common semantic core of the diverse readings associated with imperfective morphology, Hacquard (2006) proposes that we may assign it no meaning at all, and treat it as a kind of ‘default morphology’, which appears in contexts where the perfective cannot occur due to its more specific semantics. Its various contextually determined readings result from the covert modal operator PROG and the covert modal quantifiers GEN and HAB, which both bind free eventuality variables with a ‘quasi-universal’ force. (See Arregui et al. 2014 for arguments against the view that imperfectives carry no specific semantic information.) This contrasts with perfective morphology that introduces an existential quantification over eventualities (see Lenci and Bertinetto 2000), and so is systematically linked only to episodic readings.

However, the universal quantifier-based accounts of genericity/habituality, just like plurality-based ones (e.g., Ferreira 2005a, 2016, see above), cannot handle generic sentences that do not require supporting instances in the actual world (see e.g., (3a). Moreover, aligning genericity/habituality with imperfectivity is also problematic (also *pace* Comrie 1976, i.a.), because it implies that perfective sentences rule out generic and habitual interpretations. However, this is not the case. Consider the following attested Italian example (the language that Lenci and Bertinetto 2000 draw their evidence from).¹

- (29) Mia madre ha sempre comprato al mercato contadino ... e molte volte mi portava con sé.
 my mother has.AUX always bought at.the market farmer ... and many times me take.IPFV with self
 ‘My mother always used to shop/shopped (PASSATO PROSSIMO/PFV) at the farmers’ market ... and many times took (IPFV) me with her.’ Italian

¹Retrieved at <https://www.facebook.com/DottFrancescaRuggiero/posts>, July 2, 2024, Dott.ssa Francesca Ruggiero, accessed December 1, 2025.

As (29) shows, perfective forms can be used in Italian sentences that have a generic/habitual interpretation, and also be combined with quantificational adverbs facilitating such an interpretation (contrary to Lenci and Bertinetto 2000). As we have seen, generic/habitual statements can be expressed by either imperfective or perfective forms. Consequently, genericity/habituality cannot be uniquely tied to imperfectivity (see also (Filip, 1993, 1994; Filip and Carlson, 1997; Filip, 2026)).

6 Conclusion

In order to advance our understanding of genericity and habituality, and their unique kind of modality, there should be increased focus on cross-linguistic semantics and typology, and the empirical foundation of genericity and habituality. Specifically, this concerns the exploration of overt markers of genericity and habituality across a broad swath of languages: namely, markers that systematically enforce only a generic (habitual) interpretation of sentences, excluding any episodic interpretations, and that are traditionally (and perhaps misleadingly) labeled as “habitual”, “iterative”, “frequentative”, “multiplicative”, and the like, and subsumed under tense and/or aspect. Upon closer and rigorous examination, and taking all the relevant data into account, it may very well turn out that among these are markers that are best analyzed as generic markers endowed with a modal (intensional) import that has been identified as characteristic of bona fide generic sentences. In-depth studies of such markers seems one of the best strategies for evaluating the extant hypotheses about the modality of generic (and habitual) sentences in linguistics, and in other fields of cognitive science.

The study of linguistic phenomena that fall under what is understood as “genericity”, and its specific modality, goes well beyond linguistics. It also is explored in other areas of cognitive science, including philosophy, psychology. Much of our knowledge about the world is conveyed by generic (and habitual) sentences expressing generalizations of various types. Generic (and habitual) sentences are cognitively preeminent and omnipresent in language, and the ability to infer generalizations from particular samples, facts, or instances is a prerequisite to forming expectations that allow us to respond appropriately to novel samples and particulars. This ability to generalize is of key importance to acquiring knowledge about the world and to reasoning about it.

The modality of generics (and habituals) that stems from their hallmark property of exception-tolerance raises fundamental questions about what we view as characteristic, typical, normal, expected, or significant, on the one hand, and what we consider as failing to conform, to be aberrant, exceptional, on the other hand. Recent work, especially in philosophy, focuses on the related normative facets of

generic sentences, including the idea that they can be carriers of ‘epistemic injustice’ towards groups of people (e.g., Rosola and Cella, 2020). For example, *Soccer players are overpaid and lazy* arguably is a generalization based on some observations, but it resists easy falsification due to its exception-tolerance (e.g., Cimpian et al., 2010; Bian and Cimpian, 2021; Gelman et al., 2010). Another possible modal source for the resistance against falsification of such generics is that they are interpreted with respect to a set of accessible possible worlds, but it is not entirely determined which set this precisely is. In this way, generic sentences can appear to express subjective attitudes, akin to so-called predicates of personal taste (Snyder 2013, Pearson 2022), and lack verifiable truth values.

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